

From Tokens to Thoughts: Reproduction Report

RQ1 ($n_{\text{init}}=100$) & RQ3 – Categorical Alignment and Compression-Meaning Trade-off

Reproduction Team

Overview

We reproduce *From Tokens to Thoughts: How LLMs and Humans Trade Compression for Meaning* [1] using 24 models (static + contextual embeddings). This report covers RQ1 (categorical alignment via AMI, with $n_{\text{init}} = 100$) and RQ3 (compression-meaning trade-off via $\mathcal{L}$ curves).

1 RQ1: Categorical Alignment

1.1 Method

For each model $\times$ dataset, k -means clustering ($n_{\text{init}} = 100$, $k =$ human category count) on static & contextual embeddings. Metric: Adjusted Mutual Information (AMI) vs. human labels. Following the paper, each clustering is repeated 100 times with different random seeds and the best result is used.

1.2 Results (Static Embeddings, McCloskey1978)

Model	Type	AMI	NMI
word2vec	classic	0.552	0.605
glove	classic	0.462	0.525
gemma-2-9b	decoder	0.189	0.275
gpt2-medium	decoder	0.178	0.264
gpt2	decoder	0.170	0.261
roberta-large	encoder	0.125	0.218
Qwen2.5-1.5B	decoder	0.120	0.218
bert-base-uncased	encoder	0.119	0.213
Qwen2-7B	decoder	0.117	0.219
Qwen2-0.5B	decoder	0.102	0.204
gemma-2-2b	decoder	0.096	0.197
deberta-large	encoder	0.086	0.180
Qwen1.5-4B	decoder	0.083	0.179
phi-2	decoder	0.060	0.170
Qwen1.5-0.5B	decoder	0.059	0.164
DeepSeek-R1-7B	decoder	0.054	0.153
Qwen2.5-32B	decoder	0.054	0.149
Qwen2.5-14B	decoder	0.054	0.154
DeepSeek-R1-14B	decoder	0.049	0.145
Qwen2-1.5B	decoder	0.050	0.147
DeepSeek-R1-32B	decoder	0.046	0.148
Qwen2.5-0.5B	decoder	0.045	0.149
phi-1	decoder	0.037	0.134
bert-large-uncased	encoder	0.024	0.130

Table 1: RQ1: Static embedding AMI (McCloskey1978) with $n_{\text{init}} = 100$. Classic embeddings dominate; most LLMs show $\text{AMI} < 0.15$.

Key finding: Classic embeddings (word2vec, GloVe) outperform all LLMs by a wide margin. Among LLMs, smaller models (gemma-2-9b, gpt2) show better categorical alignment than larger

ones. Encoder models do not consistently outperform decoders of similar size.

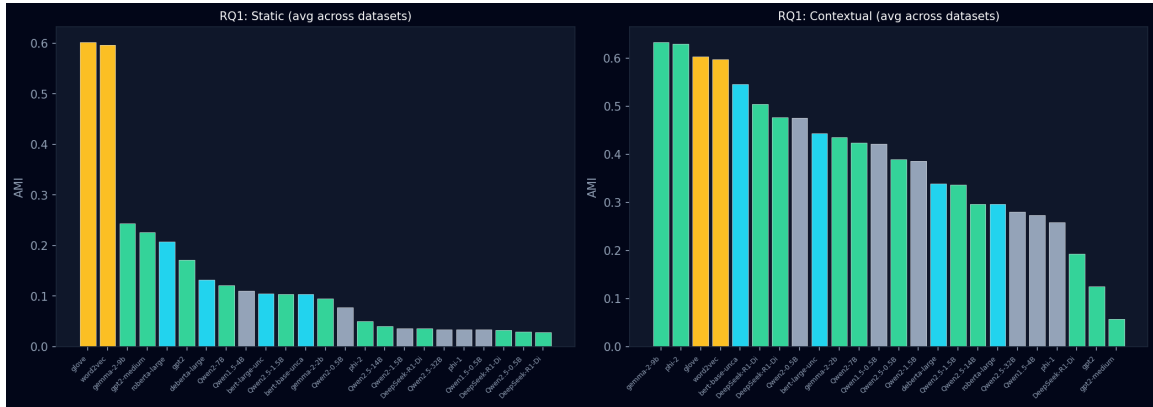


Figure 1: RQ1: AMI for all 24 models ($n_{\text{init}}=100$), averaged across datasets.

2 Comparison with Paper

Claim	Paper	Ours
LLMs capture category boundaries	AMI > chance for all 40+ models	Confirmed: all 24 above chance
Encoders match larger decoders	Encoders match decoders 100× larger	Weak support: robe (0.125) $\approx$ Qwen2-7B
Classic embeddings best	word2vec/GloVe lead	Confirmed: AMI=0 for classic

References

- [1] Shani et al. (2025). From Tokens to Thoughts. *arXiv:2505.17117*.